

RAttAGS: X-ray Attenuation-Aligned 3D Gaussian Splatting for Sparse Tomographic View Synthesis

Abstract

*3D Gaussian Splatting has been extended to X-ray novel view synthesis, yet existing methods retain the adaptive density control from natural-image 3DGS without modification. Under sparse CT acquisition, these rules cause Gaussians to accumulate near high-contrast anatomical boundaries while interior tissues that determine X-ray attenuation remain under-represented. We present **RAttAGS**, an X-ray Attenuation-Aligned 3D Gaussian Splatting framework that corrects this structural imbalance by redesigning the adaptive density control pipeline. RAttAGS introduces three components: a support-profile seeding strategy that initializes Gaussians from the coarse attenuation support estimated by FDK reconstruction; a geometry-aware pruning step that removes Gaussians concentrated near anatomical boundaries; and an adaptive density modulation step that applies position-dependent density refinement throughout training. Experiments on five organs under a sparse-view CT protocol demonstrate that RAttAGS achieves state-of-the-art novel view synthesis quality against six representative baselines from both visible-light sparse-view synthesis and X-ray CT methods.*

1. Introduction

Computed tomography (CT) acquires multi-angle X-ray attenuation measurements to reconstruct an object’s internal structure, and is indispensable in clinical diagnosis, image-guided intervention, and industrial inspection [FDK84; AK84; WYD08]. Reducing the number of acquired views directly lowers radiation dose, shortens scan time, and broadens applicability to patients who cannot sustain long acquisitions [LMC*23]. In this work, we study *Sparse*

Tomographic View Synthesis: from only a handful of X-ray projections, the model must predict the line-integrated attenuation at unseen acquisition angles, so that each synthesized projection remains consistent with the Beer–Lambert imaging geometry. A reliable solution supplies auxiliary views for anatomical localization and intraoperative planning where conventional analytical reconstruction breaks down [GKC*21].

Visible-light imaging captures surfaces through

reflection and scattering, whereas X-ray imaging reveals internal structures through penetration and attenuation. These two fundamentally different physical processes that dictate where representation capacity should reside. In visible-light 3DGS, a pixel is formed by front-to-back α -compositing [KKLD*23; Max02]: transmittance decays multiplicatively, so for opaque scenes the integrand is dominated by Gaussians near the first-hit surface, a transmittance-induced concentration consistent with surface-aligned 3DGS variants [GL24; HYC*24]. In X-ray imaging, the detector reading follows the Beer–Lambert law [KS01]: after taking the logarithm, the recorded quantity becomes a linear, order-independent line integral of the attenuation coefficient, with no transmittance decay and no occlusion. Every Gaussian on the path therefore contributes on equal footing, and capacity must be distributed *along the entire ray path*. Existing X-ray Gaussian methods have correctly replaced α -compositing with a radiative line-integral renderer [CLW*24; ZLC*24; FJJ*25; WLG*25], and sparse-view 3DGS pipelines further compensate for missing observations through depth or co-regularization priors [ZFJW24; ZLY*24; LZB*24]. Yet all of them inherit an adaptive density control strategy originally designed for surface-centric α -compositing [KKLD*23; RPK24].

Crucially, the issue is not gradient-driven densification itself; splitting or cloning Gaussians where the projection-error gradient is large is sound, and our method retains this mechanism. The issue is that under a line-integral renderer, the residual gradient is still concentrated at high-attenuation-contrast interfaces such as bone–soft-tissue boundaries. A naïve port therefore keeps spawning Gaussians at anatomical boundaries even though the rendering equation asks for capacity *along the path*. Under sparse views,

this bias manifests as a structural *capacity misallocation*: Gaussians accumulate at boundaries while interior path segments that determine the attenuation integral remain under-represented, as illustrated in Fig. 1. Fixing this requires *augmenting*, not discarding, gradient-driven densification with attenuation-aligned decisions about where to seed, what to prune, and how to refine.

We propose **RAttAGS**, an X-ray Attenuation-Aligned 3D Gaussian Splatting framework for Sparse Tomographic View Synthesis. **RAttAGS** keeps the gradient-driven backbone of standard 3DGS and adds three attenuation-aligned stages that act on *where* Gaussians are seeded, retained, and refined. Support-Profile Seeding (SPS) initializes Gaussians from a coarse FDK attenuation support [FDK84] so that they populate physically meaningful X-ray paths; Geometry-aware Pruning (GAP) recycles Gaussians that accumulate near high-contrast anatomical interfaces, draining boundary-driven redundancy at its source; and Adaptive Density Modulation (ADM) applies position-dependent online refinement so that retained Gaussians carry interior attenuation rather than uniformly inflated density.

Our main contributions are summarized as follows:

- We identify *capacity misallocation* as the core failure mode when sparse-view X-ray Gaussian methods inherit visible-light adaptive density control under a line-integral renderer.
- We propose **RAttAGS**, an X-ray Attenuation-Aligned 3D Gaussian Splatting framework that augments gradient-driven densification with three attenuation-aligned stages (SPS for path-anchored initialization, GAP for boundary-redundancy recovery, and ADM for position-dependent density refinement) without modifying the X-ray renderer.

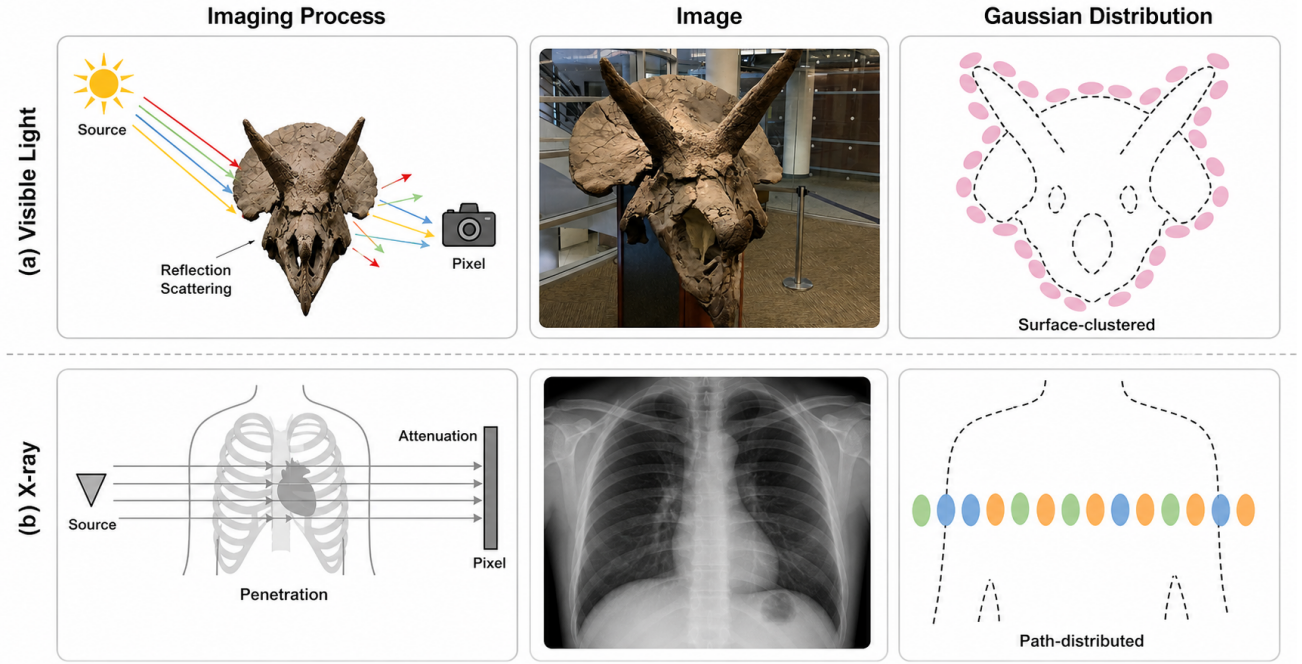


Figure 1: Visible-light vs. X-ray Gaussian rendering and the capacity gap they impose. (a) Visible-light α -compositing concentrates capacity on visible surfaces, where front Gaussians occlude back ones. (b) X-ray line integrals weight every Gaussian along the ray equally, so naïve gradient-driven densification still clusters at attenuation boundaries; **RAttAGS** realigns the density control strategy through SPS, GAP, and ADM to anchor capacity along the path.

- Experiments on multi-organ sparse-view CT show that **RAttAGS** delivers consistent gains over state-of-the-art baselines; ablations confirm that SPS, GAP, and ADM each contribute complementary capacity reallocation.

2. Related Work

2.1. Gaussian Splatting

3D Gaussian Splatting (3DGS) represents a scene with a set of anisotropic Gaussians and renders novel views in real time through differentiable tile-based rasterization [KKLD*23]. Building on this explicit representation, subsequent work has expanded along several directions, including surface-aligned variants [HYC*24; GL24], dynamic and deformable

extensions, dense visual SLAM, and compactness redesigns. Across all these directions, the underlying adaptive density control remains the gradient-driven clone-and-split strategy together with spherical-harmonics view-dependent color, both of which are tailored to dense-view, surface-reflectance scenes. When either assumption is violated (inputs become sparse, or the imaging model switches from surface radiance to line-integral attenuation), this strategy starts to misallocate Gaussian capacity, and the next two subsections examine these two regimes in turn.

2.2. Sparse View Synthesis

Neural Radiance Fields (NeRF) recover a continuous volumetric field through coordinate MLPs and volume rendering [MST*21], and degrade notice-

ably when only a handful of input views are available. NeRF-side sparse-view methods approach this limitation from complementary angles, each adding a different external prior: depth supervision from sparse structure-from-motion points [DLZR22] and geometric regularization with frequency-domain annealing [NBM*22]. Sparse-view 3DGS work explores a similarly diverse set of perspectives on the explicit Gaussian set: FSGS attributes degradation to insufficient coverage and augments densification with proximity-guided unpooling and depth priors [ZFJW24]; CoR-GS treats it as unstable optimization and adds dual-field co-regularization with co-pruning [ZLY*24]; DNGaussian injects geometry through depth normalization [LZB*24]; and DropGaussian counters overfitting via stochastic Gaussian dropout [PRK25]. A complementary feed-forward line further moves geometry completion into large-scale pretraining [CLTS24]. These different angles all yield empirical gains, which suggests that sparse-view degradation admits multiple compatible explanations rather than a single root cause. The next subsection turns to how that assumption itself is challenged under X-ray line integration.

2.3. Tomographic Novel View Synthesis

Sparse-view tomographic novel view synthesis aims to recover unseen X-ray projections, and the underlying density volume, from a small number of input projections [WYD08; GKC*21], in contrast to analytical (FDK) and iterative (SART) reconstructions that either suffer severe streak artifacts under sparse views or require many iterations [FDK84; AK84]. Learning-based methods fall into two families with complementary entry points. Implicit methods adapt neural attenuation fields to X-ray imaging, including self-supervised sinogram synthesis, hash-encoded acceleration, transformer-based structural modeling, or population-level encoder-decoder infer-

ence [ZIL*21; ZZL22; CWY*24; LFL*24]. Explicit methods redesign Gaussians around X-ray attenuation, covering radiative kernels with X-ray rasterization, integration-bias correction with differentiable voxelization, material-aware ellipsoids with explicit segment-length physics, and discretized voxel representations [CLW*24; ZLC*24; FJJ*25; WLG*25]. Across both families, improvements have concentrated on the imaging equation or the kernel itself rather than on the density control strategy that decides where Gaussians accumulate, leaving sparse-view CT still bottlenecked by capacity misallocation inherited from visible-light densification. **RAttAGS** addresses this density control strategy directly, with SPS, GAP, and ADM aligning initialization, pruning, and refinement to the X-ray attenuation path.

3. Method

3.1. Preliminary

3.1.1. X-ray Attenuation Imaging

An X-ray detector pixel records the residual energy of a single ray after traversing the object, governed by the Beer–Lambert law. For a ray $\mathbf{r}$ with source intensity I_0 and attenuation field $\mu(\mathbf{x})$, the standard practice is to work in the logarithmic domain [KS01], so that each pixel becomes a line integral of attenuation along the ray:

$$I(\mathbf{r}) = -\log \frac{I'(\mathbf{r})}{I_0} = \int_{\mathbf{r}} \mu(\mathbf{x}) d\mathbf{l}. \quad (1)$$

Unlike α -compositing in visible-light rendering, this integral is linear and order-independent: every point along $\mathbf{r}$ contributes additively, with no occlusion between front and back materials.

3.1.2. Radiative Gaussian Backbone

We build upon the R2-Gaussian framework [ZLC*24], which represents the scanned object with a set of radiative Gaussians $\mathcal{G} = \{G_i\}_{i=1}^N$.

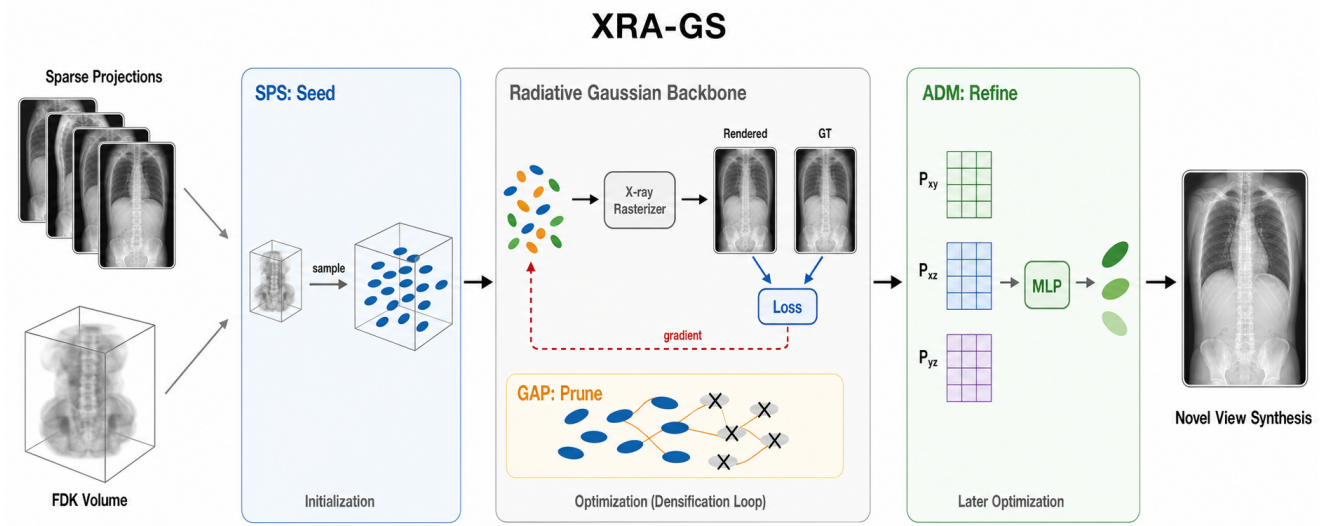


Figure 2: Overall framework of RAttAGS. Sparse X-ray projections together with a coarse FDK reconstruction enter a radiative Gaussian backbone whose adaptive density control is reorganized by SPS (seed), GAP (prune), and ADM (refine). The three modules act at initialization, inside the densification loop, and during later optimization, respectively, leaving inference unchanged.

Each Gaussian G_i is parameterized by a center $\mathbf{p}_i$, a covariance Σ_i , and a non-negative density ρ_i , defining a local density field

$$G_i(\mathbf{x}) = \rho_i \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{p}_i)^\top \Sigma_i^{-1}(\mathbf{x} - \mathbf{p}_i)\right), \quad (2)$$

and the global attenuation field is the sum $\mu(\mathbf{x}) = \sum_{i=1}^N G_i(\mathbf{x})$. Substituting this representation into Eq. (1) admits a differentiable X-ray rasterizer $\mathcal{R}$, so that for any acquisition geometry π_v a projection can be rendered as

$$\hat{I}_v = \mathcal{R}(\mathcal{G}; \pi_v). \quad (3)$$

3.1.3. Problem Statement

Given a sparse-view observation set $\mathcal{P} = \{(I_v, \pi_v)\}_{v=1}^V$ and a coarse FDK reconstruction V_{FDK} , our goal is to learn $\mathcal{G}$ in a per-case optimization setting so that target projections at unseen angles can be synthesized from only a few observed projections. The radiative backbone has already

aligned the renderer with X-ray physics; what remains misaligned in this regime is the density control strategy that shapes $\mathcal{G}$ during training, where adaptive density control continues to concentrate Gaussians at high-gradient anatomical boundaries. **RAttAGS** realigns this density control strategy through three modules introduced below, while leaving the rasterizer and the differentiable voxelizer of R2-Gaussian [ZLC*24] unchanged.

3.2. Overall Framework

As illustrated in Figure 2, **RAttAGS** instantiates the attenuation-aligned density control pipeline as three modules acting at three distinct stages of optimization: SPS replaces standard initialization by seeding Gaussians along the FDK attenuation support; GAP acts inside the densification loop and reclaims locally redundant Gaussians produced by gradient-driven growth; ADM acts on retained Gaus-

sians and refines their density values using position-dependent spatial context. At inference, the rasterizer remains identical to the radiative backbone, so the added cost is concentrated in training.

3.3. Support-Profile Seeding (SPS)

Initialization strongly conditions the optimization trajectory of radiative Gaussians, yet standard 3DGS initializers transfer poorly to X-ray tomography: uniform random sampling ignores the strong contrast between air, soft tissue, and bone, while SfM point-cloud initialization is unstable under gray-value overlap and weak local texture in X-ray projections [CLW*24]. X-Field [FJJ*25] addresses this by progressively refining a coarse CT estimate through a CGLS-SART-TV pipeline before initializing ellipsoids at the reconstructed voxel centers with random covariance and attenuation attributes. While effective, this multi-stage iterative reconstruction adds considerable overhead. We observe that a single-pass FDK volume V_{FDK} , despite streak artifacts in fine detail, already provides reliable foreground support and low-frequency attenuation layout [WCS*22; LMC*23; LFL*25; ZLC*24], which suffices as a sampling prior for the subsequent density control stages (GAP, ADM) to refine.

Concretely, over the voxel grid Ω underlying V_{FDK} we draw N seed positions $\{\mathbf{x}_i\}_{i=1}^N$ from a density-weighted mixture distribution

$$q(\mathbf{x}) = \alpha \frac{1}{|\Omega|} + (1-\alpha) \frac{V_{\text{FDK}}(\mathbf{x})^\gamma}{Z}, \quad Z = \sum_{\mathbf{x}' \in \Omega} V_{\text{FDK}}(\mathbf{x}')^\gamma, \quad (4)$$

that interpolates between a uniform prior (weight α) and a power-shaped attenuation profile (exponent γ). Each sampled position $\mathbf{x}_i$ then defines a Gaussian G_i with attributes initialized as:

$$\boldsymbol{\mu}_i = \mathbf{x}_i, \quad \sigma_i = V_{\text{FDK}}(\mathbf{x}_i), \quad s_i = \frac{1}{K} \sum_{j \in \mathcal{N}_K(i)} \|\mathbf{x}_i - \mathbf{x}_j\|_2, \quad (5)$$

where $\boldsymbol{\mu}_i$ is the center position, σ_i the initial attenuation coefficient queried directly from the FDK volume, and s_i the isotropic scale set by the mean distance to the K -nearest seed neighbors $\mathcal{N}_K(i)$. The mixture form is critical: density weighting alone would collapse capacity onto the brightest bony structures, while the uniform term retains global coverage so that under-attenuating but anatomically relevant soft-tissue regions remain populated.

3.4. Geometry-aware Pruning (GAP)

Inside the radiative backbone, projection-error gradients are systematically larger near bone-soft-tissue boundaries, so standard gradient-driven densification keeps duplicating Gaussians around the same high-contrast surfaces. Large gradients in this regime do not always imply missing coverage: they can also reflect that an already-saturated boundary is being represented redundantly by neighboring Gaussians. Rather than adding more Gaussians in sparse regions in the spirit of FSGS [ZFJW24], GAP acts in the opposite direction and reclaims locally crowded Gaussians during densification.

Following the proximity-graph construction of FSGS [ZFJW24], we connect each Gaussian G_i to its K nearest neighbors $\mathcal{N}_K(i)$ in position space and compute a proximity score as the mean neighbor distance:

$$d_i = \frac{1}{K} \sum_{j \in \mathcal{N}_K(i)} \|\mathbf{p}_i - \mathbf{p}_j\|_2. \quad (6)$$

A small d_i indicates that G_i sits in a locally crowded region. We combine the proximity score with $\bar{g}_i$, the accumulated positional-gradient magnitude of G_i over the current densification cycle, and flag a Gaussian as a redundancy candidate only when it is simultaneously locally crowded and recently inactive:

$$c_i = \mathbf{1}[d_i < \tau \wedge \bar{g}_i < \delta], \quad (7)$$

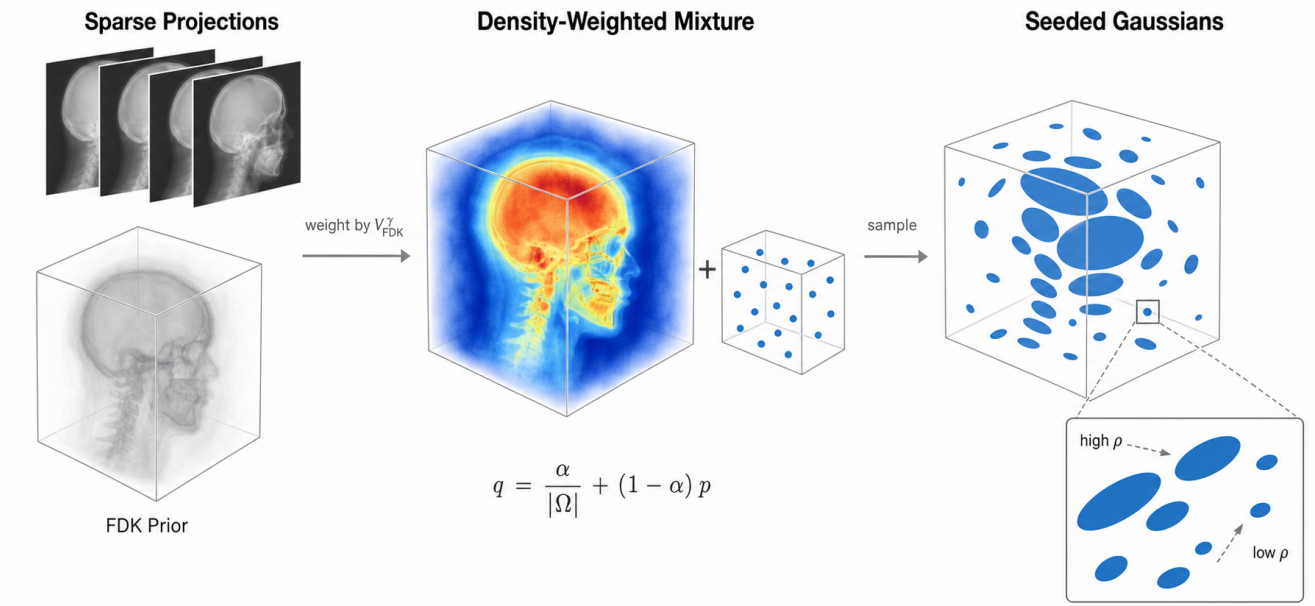


Figure 3: Support-Profile Seeding (SPS). From the coarse FDK volume, SPS builds a density-weighted mixture distribution that interpolates between a uniform prior and the attenuation profile. Sampling from this mixture seeds Gaussians along the anatomical support before any optimization step.

where $c_i = 1$ marks a prune candidate. GAP removes a bounded fraction of the flagged Gaussians per cycle and slightly contracts the covariance of their retained neighbors to avoid creating local support holes. The dual criterion distinguishes GAP from purely density-based pruning: gradient activity prevents deleting Gaussians that are still learning a boundary, while proximity prevents over-densified regions from continuously monopolizing capacity.

3.5. Adaptive Density Modulation (ADM)

Even after path-anchored initialization and redundancy control, density values across anatomy remain heterogeneous: bone exhibits higher attenuation, air-filled regions should stay near zero, and soft tissue requires smoother transitions. A per-Gaussian gradient step updates each density value independently, making it difficult to capture the spatial neighbor-

hood context needed to coordinate density allocation across these distinct tissue regimes. A straightforward remedy would be to employ a coordinate MLP to directly parameterize position-dependent density corrections, but such a network is prone to overfitting under sparse views where only a handful of projections constrain the entire volume. A dense 3D feature grid avoids this issue yet incurs memory that scales cubically with resolution. Inspired by the factorized plane representation of K-Planes [FMW*23], ADM decomposes the 3D volume into three orthogonal 2D feature planes, reducing memory to quadratic growth while preserving effective spatial neighborhood encoding. From these plane features, ADM decodes a position-dependent density correction for each Gaussian without modifying its center or covariance.

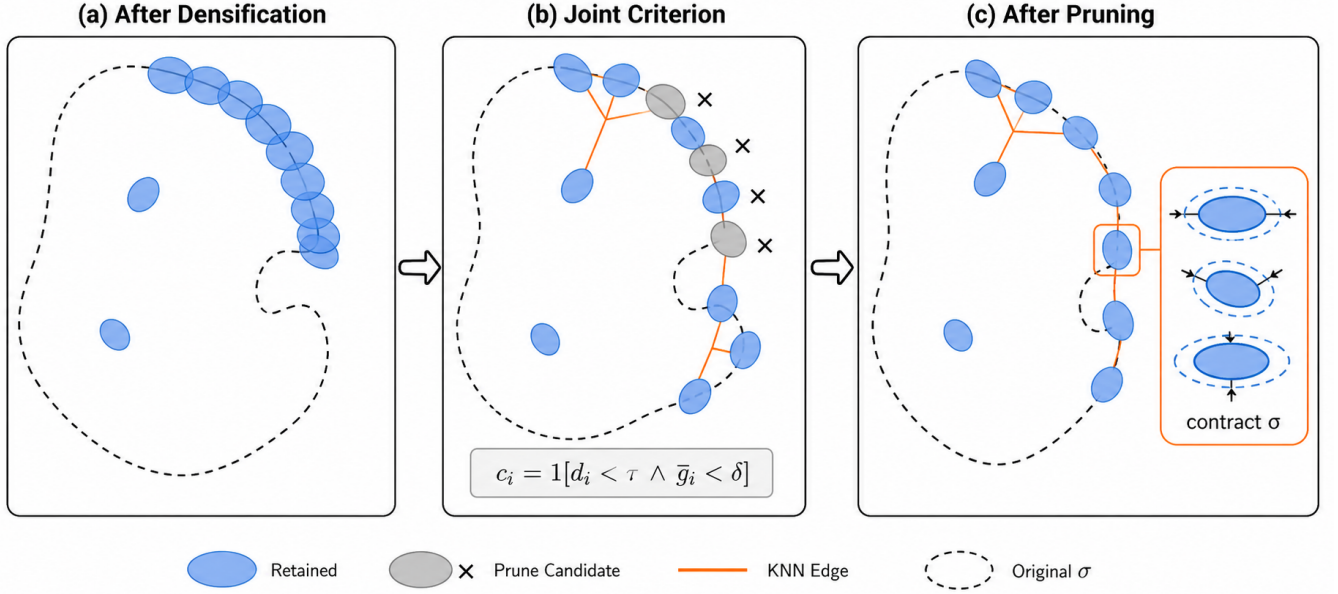


Figure 4: Geometry-aware Pruning (GAP). After gradient-driven densification, Gaussians cluster around high-contrast boundaries. GAP flags a Gaussian as a redundancy candidate only when it is both locally crowded ($d_i < \tau$) and recently inactive ($\bar{g}_i < \delta$), so that capacity is reclaimed from saturated boundaries without removing Gaussians that are still learning structure.

3.5.1. Spatial Feature Encoding

Given a Gaussian center $\mathbf{x}$, ADM projects it onto three orthogonal planes $\{P_{uv}\}_{uv \in \{xy, xz, yz\}}$, extracts per-plane features via bilinear interpolation at the corresponding 2D projections $\Pi_{uv}(\mathbf{x})$, and concatenates them into a joint spatial descriptor:

$$f_{uv}(\mathbf{x}) = \mathcal{B}(P_{uv}, \Pi_{uv}(\mathbf{x})), \quad uv \in \{xy, xz, yz\}, \quad (8)$$

$$F(\mathbf{x}) = [f_{xy}(\mathbf{x}); f_{xz}(\mathbf{x}); f_{yz}(\mathbf{x})], \quad (9)$$

where $\mathcal{B}(\cdot)$ denotes bilinear sampling. The three-plane factorization implicitly encodes 3D spatial context: each sampled feature captures a 2D slice of the neighborhood, and their concatenation provides cross-plane spatial constraints that a single plane cannot express alone.

3.5.2. Density Modulation

The concatenated feature $F(\mathbf{x})$ is fed into a lightweight dual-head MLP h that outputs a density offset $\Delta\sigma(\mathbf{x}) \in [-1, 1]$ and a confidence gate $g(\mathbf{x}) \in [0, 1]$. To prevent ADM from degenerating into a global density scaling factor, we subtract the batch mean $\overline{\Delta\sigma}$ before applying a multiplicative modulation to the base density:

$$\rho_{\text{final}}(\mathbf{x}) = \rho_{\text{base}}(\mathbf{x}) \cdot \left(1 + g(\mathbf{x}) \cdot (\Delta\sigma(\mathbf{x}) - \overline{\Delta\sigma})\right). \quad (10)$$

The plane features and the MLP parameters are trained jointly with the backbone Gaussians from the start of optimization under the rendering loss alone, with no extra density supervision. This joint training allows ADM to impose spatial regularization on capacity allocation while the backbone is still learning the geometric layout, rather than correcting a

frozen density distribution post hoc. Batch centering ensures that $\Delta\sigma$ acts as a relative correction: ADM amplifies density where the local attenuation evidence warrants finer detail and suppresses modulation where evidence is insufficient.

3.6. Training Loss

RAttAGS is supervised end-to-end on the observed views by a single objective combining projection-domain reconstruction with regularization on the volumetric density and the K-Planes features:

$$\mathcal{L} = \mathcal{L}_1 + \lambda_d \mathcal{L}_{\text{dssim}} + \lambda_v \mathcal{L}_{3\text{DTV}} + \lambda_p \mathcal{L}_{\text{pTV}}. \quad (11)$$

The first two terms are the standard L_1 and D -SSIM losses between rendered and observed projections; $\mathcal{L}_{3\text{DTV}}$ is a total-variation prior on the volumetric density retrieved by the differentiable voxelizer of R2-Gaussian, and $\mathcal{L}_{\text{pTV}}$ is a planar total-variation prior on the K-Planes features used by ADM. The K-Planes features and the MLP parameters of ADM are jointly updated with the backbone Gaussian parameters under $\mathcal{L}$, so density modulation is learned online rather than fixed in advance.

4. Experiments

4.1. Experimental Settings

Dataset. Following NAF [ZZL22] and X-Gaussian [CLW*24], we adopt the public datasets of human organ CTs, *i.e.*, LIDC-IDRI and the open scientific visualization dataset, to evaluate our method. The test scenes include Chest, Foot, Head, Abdomen, and Pancreas. We adopt the open-source tomographic toolbox TIGRE to capture 100 projections with 3% noise in the range of $0 \sim 180^\circ$. In the sparse-view NVS task, we retain only 2, 3, or 4 projections for training, and the remaining projections are used for testing.

Baselines. We compare **RAttAGS** with

six state-of-the-art methods from two categories. *X-ray/CT reconstruction methods:* X-Gaussian [CLW*24] [ECCV'24] introduces radiative Gaussian splatting for X-ray NVS; R2-Gaussian [ZLC*24] [NeurIPS'24] rectifies the integration bias in 3DGS and develops a differentiable voxelizer for tomographic reconstruction; and X-Field [FJJ*25] [NeurIPS'25] employs physically informed ellipsoids with material-aware optimization. *Natural-scene sparse-view 3DGS methods:* CoR-GS [ZLY*24] [ECCV'24] introduces co-regularization to stabilize sparse-view training; DNGaussian [LZB*24] [CVPR'24] injects depth normalization for geometric guidance; and FSGS [ZFJW24] [ECCV'24] augments densification with proximity-guided unpooling. All methods are evaluated under the same sparse-view protocol on the same dataset.

Metrics. We adopt peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) to evaluate the performance. Both metrics are computed on $[0, 1]$ -normalized projections, averaged over all held-out views of each scene and then over the five organs.

Implementation Details. Our **RAttAGS** is implemented in PyTorch and CUDA. The model is trained with the Adam optimizer for 3×10^4 iterations. SPS uses mixture coefficient $\alpha = 0.2$, power exponent $\gamma = 1.0$, and 50K initial Gaussians. GAP uses $K = 5$ nearest neighbors, proximity threshold $\tau = 0.015$, maximum pruning ratio $\beta_{\text{prune}} = 2\%$, and gradient threshold $\delta = 0.0002$. ADM uses K-Planes at resolution 64×32 and is jointly trained from the start of optimization. All experiments are conducted on a single NVIDIA RTX A6000 GPU.

Adaptive Density Modulation (ADM)

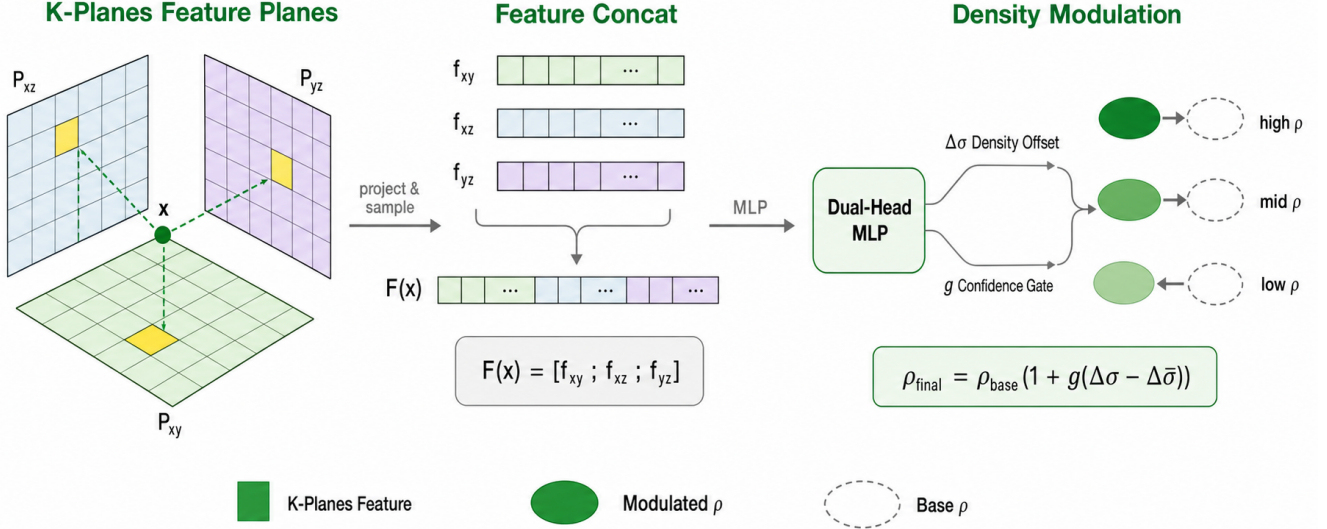


Figure 5: Adaptive Density Modulation (ADM). For each Gaussian center, three orthogonal K-Planes feature planes are bilinearly sampled and concatenated; a dual-head MLP outputs a density offset and a confidence gate, which together apply a batch-centered relative correction $\rho_{\text{final}} = \rho_{\text{base}}(1 + g(\Delta\sigma - \Delta\bar{\sigma}))$.

Table 1: PSNR comparison on the novel view synthesis task across five organs under 2-, 3-, and 4-view sparse settings. We colorize the best, second-best, and third-best results.

Method	Chest			Head			Abdomen			Foot			Pancreas			Average		
	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view
<i>Natural-scene sparse-view 3DGS methods</i>																		
CoR-GS [ZLY*24] [ECCV*24]	14.96	18.55	20.29	16.24	20.36	23.40	18.50	22.45	23.03	20.17	25.53	26.83	16.15	23.25	23.96	17.20	22.03	23.50
DNGaussian [LZB*24] [CVPR*24]	18.84	20.55	20.99	15.72	18.04	16.24	14.30	16.02	15.76	19.33	23.51	24.72	19.06	22.43	25.47	17.45	20.11	20.64
FSGS [ZFJW24] [ECCV*24]	17.86	19.99	21.03	19.43	21.53	24.91	18.75	23.34	23.63	22.17	25.37	27.08	23.03	25.27	26.22	20.25	23.10	24.57
<i>X-ray/CT reconstruction methods</i>																		
X-Gaussian [CLW*24] [ECCV*24]	17.79	20.29	20.90	19.62	21.52	24.46	18.93	23.54	23.85	22.34	25.48	26.92	23.21	25.12	25.55	20.38	23.19	24.34
R2-Gaussian [ZLC*24] [NeurIPS*24]	21.05	26.22	25.47	23.54	26.60	28.66	24.85	29.26	30.56	19.39	28.48	30.04	17.83	28.61	30.90	21.33	27.83	29.13
X-Field [FJJ*25] [NeurIPS*25]	19.57	23.94	24.45	22.49	25.06	26.78	22.20	24.87	27.19	20.45	26.03	27.25	18.12	24.94	26.36	20.57	24.97	26.41
RAttAGS (Ours)	21.22	26.63	26.03	24.26	26.65	29.68	25.26	29.55	30.90	20.82	28.64	29.95	22.99	29.37	30.95	22.91	28.17	29.50

4.2. Comparison with State-of-the-Art Methods

Discussion on NVS Quantitative Results. Tables 1 and 2 report the PSNR and SSIM comparisons under the 2-view, 3-view, and 4-view sparse settings. **RAttAGS** consistently achieves the best average PSNR and SSIM across all three sparsity lev-

els, ranking first in 12 out of 15 per-organ PSNR entries and 9 out of 15 SSIM entries. The natural-scene sparse-view 3DGS methods (CoR-GS, DNGaussian, FSGS) struggle under X-ray imaging, as their densification strategies and geometric priors are designed for visible-light reflection and scattering rather than X-ray penetration and attenuation, resulting in sub-

stantially lower scores across all organs. Among the X-ray-specific methods, X-Gaussian achieves reasonable PSNR on individual organs (*e.g.*, Foot and Pancreas under 2-view) but lacks integration-bias correction, limiting its overall performance. R2-Gaussian provides strong results through its rectified rasterization and differentiable voxelizer, serving as the strongest baseline. X-Field, despite its physically informed ellipsoid representation, shows limited effectiveness under highly sparse views: its multi-stage iterative initialization degrades when only 2–4 projections are available, producing the lowest SSIM among all methods at 2-view (0.7168) and PSNR consistently below R2-Gaussian by 2–3 dB across all view budgets. Our **RAttAGS** surpasses R2-Gaussian by +1.58 dB PSNR under 2-view, where capacity misallocation is most damaging; the advantage persists at 3-view (+0.34 dB) and 4-view (+0.37 dB).

Discussion on NVS Qualitative Results.

Figure 6 presents visual comparisons on four representative organs. DNGaussian produces severe streak and speckle artifacts across all organs: the Chest scene shows heavy noise overlaying the rib structures, the Head scene exhibits granular noise that obscures cranial details, and the Pancreas scene displays pronounced vertical streak artifacts. R2-Gaussian reconstructs the overall anatomical structure substantially better, but still exhibits blurring at bone–soft-tissue boundaries, particularly visible in the zoomed-in regions of the Foot and Pancreas scenes. X-Field introduces the most prominent blurring among the X-ray-specific methods: fine bone details are nearly lost in the Foot scene, internal cranial structures are unrecoverable in the Head scene where horizontal banding artifacts further degrade quality, and the Pancreas anatomy becomes difficult to discern. These visual observations are consistent with its lowest SSIM scores in the quantitative evalua-

tion. Our **RAttAGS** effectively mitigates these artifacts by reallocating Gaussian capacity from over-densified boundaries to interior attenuation-relevant structures, recovering clearer rib boundaries in the Chest, sharper bone edges in the Foot, smoother cranial transitions in the Head, and better-preserved soft-tissue detail in the Pancreas.

4.3. Ablation Study

Component Analysis. We conduct a break-down ablation experiment to study the effect of each proposed module. We adopt the R2-Gaussian backbone as the baseline model. The results are listed in Table 3. We can observe from the table: **(i)** When adding SPS for initialization, the performance yields improvement especially under the most under-determined 2-view regime, establishing that path-anchored initialization from the coarse FDK attenuation support provides a meaningful head start for optimization. **(ii)** Adding GAP yields a consistent PSNR gain by reclaiming capacity from over-densified boundary regions, and contributes the highest SSIM among the individual modules, confirming that geometry-aware pruning effectively corrects the capacity misallocation inherited from visible-light densification. **(iii)** Adding ADM provides further refinement through position-dependent density correction, particularly benefiting scenes with heterogeneous attenuation profiles. The full **RAttAGS** framework, combining all three modules, achieves the best performance across all view counts.

Parameter Analysis. We further investigate the sensitivity of one representative hyperparameter from each module in Table 3. **(i)** The SPS mixture coefficient α balances the FDK-derived attenuation support against random initialization; $\alpha = 0.2$ offers sufficient path-anchored guidance without over-constraining the initial distribution, while

Table 2: SSIM comparison on the novel view synthesis task across five organs under 2-, 3-, and 4-view sparse settings. We colorize the **best**, **second-best**, and **third-best** results.

Method	Chest			Head			Abdomen			Foot			Pancreas			Average		
	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view
<i>Natural-scene sparse-view 3DGS methods</i>																		
CoR-GS [ZLY*24] [ECCV'24]	0.5754	0.7077	0.7636	0.7993	0.8644	0.9088	0.8645	0.9089	0.9269	0.6913	0.8511	0.8767	0.8088	0.8974	0.9084	0.7479	0.8459	0.8769
DNGaussian [LZB*24] [CVPR'24]	0.6675	0.7500	0.7791	0.7296	0.8097	0.7924	0.7663	0.8258	0.8486	0.7801	0.8569	0.8666	0.8082	0.8848	0.8982	0.7503	0.8254	0.8370
FSGS [ZFWJ24] [ECCV'24]	0.6451	0.7429	0.7729	0.8334	0.8732	0.9105	0.8580	0.9068	0.9276	0.6843	0.8500	0.8760	0.8564	0.9031	0.9148	0.7754	0.8552	0.8804
<i>X-ray/CT reconstruction methods</i>																		
X-Gaussian [CLW*24] [ECCV'24]	0.6488	0.7427	0.7689	0.8347	0.8697	0.9079	0.8595	0.9077	0.9257	0.6882	0.8481	0.8740	0.8502	0.9006	0.9124	0.7763	0.8538	0.8778
R2-Gaussian [ZLC*24] [NeurIPS'24]	0.7110	0.8367	0.8546	0.8718	0.9223	0.9550	0.9028	0.9363	0.9617	0.6722	0.8977	0.9130	0.8122	0.9199	0.9360	0.7940	0.9026	0.9241
X-Field [FJJ*25] [NeurIPS'25]	0.6843	0.8090	0.8255	0.8413	0.8878	0.9089	0.7190	0.8632	0.9103	0.6522	0.8405	0.8556	0.6871	0.8474	0.9004	0.7168	0.8496	0.8801
RAttAGS (Ours)	0.7065	0.8390	0.8586	0.8694	0.9176	0.9518	0.9068	0.9374	0.9627	0.6910	0.8993	0.9149	0.8238	0.9247	0.9361	0.7995	0.9036	0.9248

pure attenuation-weighted sampling ($\alpha = 0.0$) drops PSNR to 28.09 and pure random initialization ($\alpha = 1.0$) to 28.06. **(ii)** The GAP proximity threshold τ controls how aggressively crowded boundary Gaussians are pruned; $\tau = 0.015$ achieves the best PSNR (28.17) and SSIM (0.9036), while a smaller $\tau = 0.005$ prunes too aggressively and drops PSNR to 27.86. **(iii)** The ADM activation iteration determines when density modulation is introduced; joint training from the start (Iter=0) achieves the highest PSNR (28.17) and SSIM (0.9036), and performance remains stable across the 5K–20K range (28.09–28.16). The adopted configuration ($\alpha = 0.2$, $\tau = 0.015$, ADM from Iter=0) achieves the best performance across all swept ranges, and the small variance across different settings demonstrates that **RAttAGS** is not sensitive to individual hyperparameter choices.

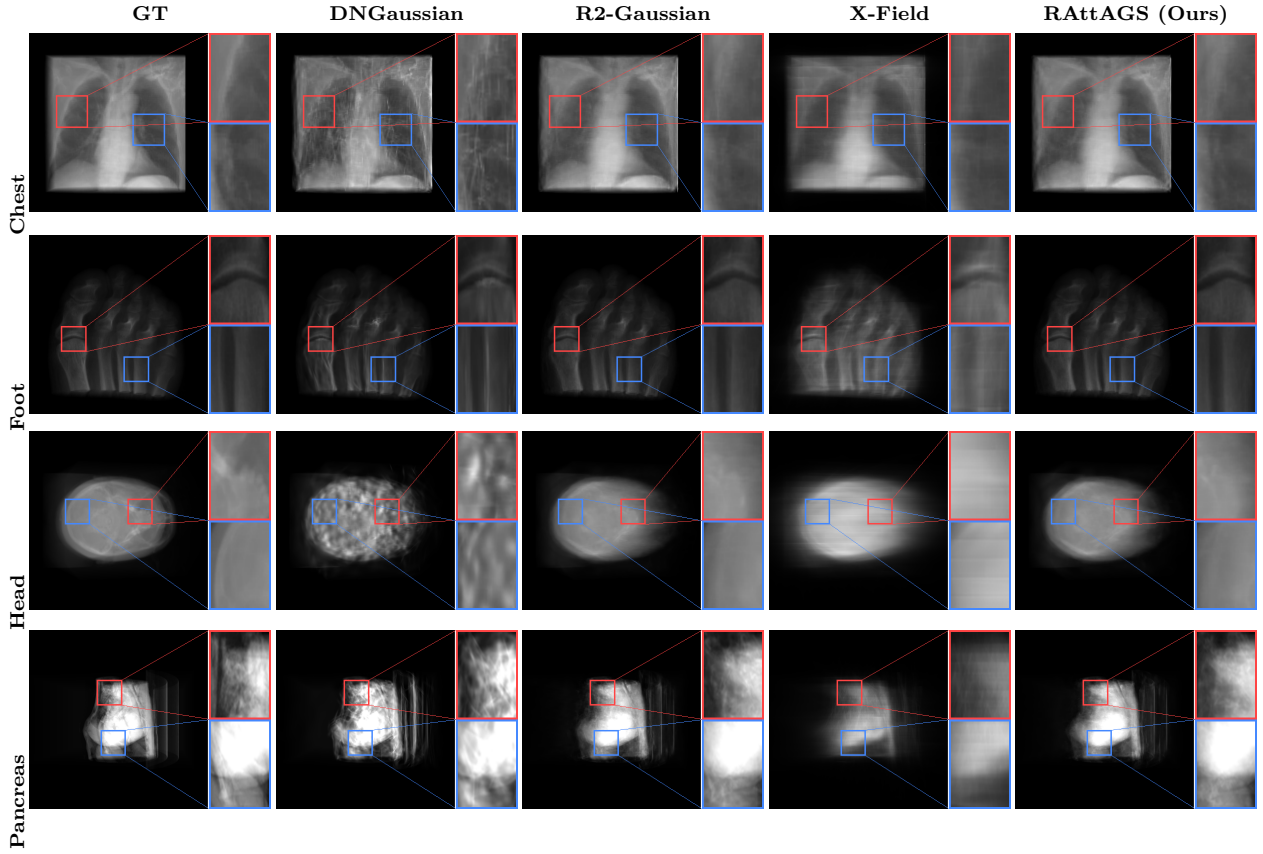


Figure 6: Qualitative comparison of novel view synthesis on four organs under the sparse-view setting. *DNGaussian produces noticeable streak and speckle artifacts; R2-Gaussian preserves the overall structure but exhibits blurring at anatomical boundaries; X-Field introduces severe blurring and banding artifacts, particularly in the Head and Foot scenes. RAttAGS recovers sharper structural details with fewer artifacts. Red and blue boxes highlight zoomed-in regions of interest.*

Table 3: *Component ablation and hyperparameter sensitivity analysis (5-organ average, 3-view). Our adopted defaults are shown in **bold**. We colorize the **best**, **second-best**, and **third-best** results.*

Configuration	PSNR $\uparrow$	SSIM $\uparrow$
<i>Component analysis (5-organ avg, 3-view)</i>		
Baseline (R2-Gaussian)	27.83	0.9026
B + SPS	28.14	0.9032
B + GAP	28.10	0.9037
B + ADM	28.06	0.9035
Full RAttAGS	28.17	0.9036
<i>SPS mixture coefficient α (5-organ avg, 3-view)</i>		
$\alpha = 0.0$	28.09	0.9036
$\alpha = 0.1$	28.13	0.9034
$\alpha = 0.2$	28.17	0.9036
$\alpha = 0.5$	28.11	0.9043
$\alpha = 1.0$	28.06	0.9031
<i>GAP proximity threshold τ (5-organ avg, 3-view)</i>		
$\tau = 0.005$	27.86	0.9030
$\tau = 0.010$	28.14	0.9031
$\tau = 0.015$	28.17	0.9036
$\tau = 0.020$	28.12	0.9031
$\tau = 0.030$	28.15	0.9030
<i>ADM activation iteration (5-organ avg, 3-view)</i>		
Iter = 0	28.17	0.9036
Iter = 5K	28.09	0.9033
Iter = 10K	28.11	0.9032
Iter = 15K	28.13	0.9034
Iter = 20K	28.16	0.9035

5. Conclusion

This paper presents **RAttAGS**, a 3DGS-based framework for sparse tomographic view synthesis. We identify a capacity misallocation problem in standard adaptive density control, where mechanisms designed for visible-light rendering concentrate Gaussians at anatomical boundaries rather than distributing them along the entire attenuation path required by X-ray line integrals. To address this, we introduce three attenuation-aligned stages that realign initialization, densification pruning, and density refinement to the X-ray attenuation path, together redirecting representation capacity toward interior structures. Experiments across diverse organs and sparse-view budgets demonstrate consistent improvements over state-of-the-art methods. Because the framework augments rather than replaces the standard densification backbone, it is in principle applicable to other transmissive modalities where line-integral physics similarly demands capacity along the entire ray path.

References

- [AK84] ANDERSEN, ANDERS H and KAK, AVINASH C. “Simultaneous algebraic reconstruction technique (SART): a superior implementation of the ART algorithm”. *Ultrasonic imaging* 6.1 (1984), 81–94 1, 4.
- [CLTS24] CHARATAN, DAVID, LI, SIZHE LESTER, TAGLIASACCHI, ANDREA, and SITZMANN, VINCENT. “pixelsplat: 3d gaussian splats from image pairs for scalable generalizable 3d reconstruction”. *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2024, 19457–19467 4.
- [CLW*24] CAI, YUANHAO, LIANG, YIXUN, WANG, JIAHAO, et al. “Radiative gaussian splatting for efficient x-ray novel view synthesis”. *European Conference on Computer Vision*. Springer. 2024, 283–299 2, 4, 6, 9, 10, 12.
- [CWY*24] CAI, YUANHAO, WANG, JIAHAO, YUILLE, ALAN, et al. “Structure-aware sparse-view x-ray 3d reconstruction”. *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2024, 11174–11183 4.
- [DLZR22] DENG, KANGLE, LIU, ANDREW, ZHU, JUN-YAN, and RAMANAN, DEVA. “Depth-supervised nerf: Fewer views and faster training for free”. *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022, 12882–12891 4.
- [FDK84] FELDKAMP, LEE A, DAVIS, LLOYD C, and KRESS, JAMES W. “Practical cone-beam algorithm”. *Journal of the Optical Society of America A* 1.6 (1984), 612–619 1, 2, 4.
- [FJJ*25] FEIRAN, WANG, JIACHEN, TAO, JUNYI, WU, et al. “X-Field: A Physically Informed Representation for 3D X-ray Reconstruction”. (2025) 2, 4, 6, 9, 10, 12.
- [FMW*23] FRIDOVICH-KEIL, SARA, MEANTI, GIACOMO, WARBURG, FREDERIK RAHBÆK, et al. “K-planes: Explicit radiance fields in space, time, and appearance”. *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2023, 12479–12488 7.
- [GKC*21] GONG, KUANG, KIM, KYUNGSANG, CUI, JIANAN, et al. “The evolution of image reconstruction in PET: from filtered back-projection to artificial intelligence”. *PET clinics* 16.4 (2021), 533–542 1, 4.
- [GL24] GUÉDON, ANTOINE and LEPETIT, VINCENT. “Sugar: Surface-aligned gaussian splatting for efficient 3d mesh reconstruction and high-quality mesh rendering”. *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2024, 5354–5363 2, 3.
- [HYC*24] HUANG, BINBIN, YU, ZEHAO, CHEN, ANPEI, et al. “2d gaussian splatting for geometrically accurate radiance fields”. *ACM SIGGRAPH 2024 conference papers*. 2024, 1–11 2, 3.
- [KKLD*23] KERBL, BERNHARD, KOPANAS, GEORGIOS, LEIMKÜHLER, THOMAS, DRETTAKIS, GEORGE, et al. “3d gaussian splatting for real-time radiance field rendering.” *ACM Trans. Graph.* 42.4 (2023), 139–1 2, 3.
- [KS01] KAK, AVINASH C and SLANEY, MALCOLM. *Principles of computerized tomographic imaging*. SIAM, 2001 2, 4.
- [LFL*24] LIU, ZHENTAO, FANG, YU, LI, CHANGJIAN, et al. “Geometry-aware attenuation learning for sparse-view CBCT reconstruction”. *IEEE Transactions on Medical Imaging* 44.2 (2024), 1083–1097 4.
- [LFL*25] LI, YINGTAI, FU, XUEMING, LI, HAN, et al. “3DGR-CT: Sparse-view CT reconstruction with a 3D Gaussian representation”. *Medical Image Analysis* 103 (2025), 103585 6.
- [LMC*23] LI, ZILONG, MA, CHENGLONG, CHEN, JIE, et al. “Learning to distill global representation for sparse-view ct”. *Proceedings of the IEEE/CVF international conference on computer vision*. 2023, 21196–21207 1, 6.
- [LZB*24] LI, JIAHE, ZHANG, JIAWEI, BAI, XIAO, et al. “Dngaussian: Optimizing sparse-view 3d gaussian radiance fields with global-local depth normalization”. *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2024, 20775–20785 2, 4, 9, 10, 12.

- [Max02] MAX, NELSON. “Optical models for direct volume rendering”. *IEEE Transactions on Visualization and Computer Graphics* 1.2 (2002), 99–108 2.
- [MST*21] MILDENHALL, BEN, SRINIVASAN, PRATUL P, TANCIK, MATTHEW, et al. “Nerf: Representing scenes as neural radiance fields for view synthesis”. *Communications of the ACM* 65.1 (2021), 99–106 3.
- [NBM*22] NIEMEYER, MICHAEL, BARRON, JONATHAN T, MILDENHALL, BEN, et al. “Regnerf: Regularizing neural radiance fields for view synthesis from sparse inputs”. *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022, 5480–5490 4.
- [PRK25] PARK, HYUNWOO, RYU, GUN, and KIM, WONJUN. “Dropgaussian: Structural regularization for sparse-view gaussian splatting”. *Proceedings of the computer vision and pattern recognition conference*. 2025, 21600–21609 4.
- [RPK24] ROTA BULÒ, SAMUEL, PORZI, LORENZO, and KONTSCIEDER, PETER. “Revising densification in gaussian splatting”. *European Conference on Computer Vision*. Springer. 2024, 347–362 2.
- [WCS*22] WANG, YANLI, CHAO, LIANYING, SHAN, WENQI, et al. “Improving the quality of sparse-view cone-beam computed tomography via reconstruction-friendly interpolation network”. *Proceedings of the Asian Conference on Computer Vision*. 2022, 2255–2269 6.
- [WLG*25] WU, SHAKAI, LU, YUXIANG, GUO, YAPAN, et al. “Discretized Gaussian representation for tomographic reconstruction”. *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2025, 25073–25082 2, 4.
- [WYD08] WANG, GE, YU, HENGYONG, and DE MAN, BRUNO. “An outlook on x-ray CT research and development”. *Medical physics* 35.3 (2008), 1051–1064 1, 4.
- [ZFJW24] ZHU, ZEHAO, FAN, ZHIWEN, JIANG, YIFAN, and WANG, ZHANGYANG. “Fsgs: Real-time few-shot view synthesis using gaussian splatting”. *European conference on computer vision*. Springer. 2024, 145–163 2, 4, 6, 9, 10, 12.
- [ZIL*21] ZANG, GUANGMING, IDOUGH, RAMZI, LI, RUI, et al. “Intratomo: self-supervised learning-based tomography via sinogram synthesis and prediction”. *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2021, 1960–1970 4.
- [ZLC*24] ZHA, RUYI, LIN, TAO JUN, CAI, YUANHAO, et al. “R²-Gaussian: Rectifying Radiative Gaussian Splatting for Tomographic Reconstruction”. *arXiv preprint arXiv:2405.20693* (2024) 2, 4–6, 9, 10, 12.
- [ZLY*24] ZHANG, JIAWEI, LI, JIAHE, YU, XIAOHAN, et al. “Cor-gs: sparse-view 3d gaussian splatting via co-regularization”. *European conference on computer vision*. Springer. 2024, 335–352 2, 4, 9, 10, 12.
- [ZZL22] ZHA, RUYI, ZHANG, YANHAO, and LI, HONGDONG. “NAF: neural attenuation fields for sparse-view CBCT reconstruction”. *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer. 2022, 442–452 4, 9.